Milestone

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Milestone is an app designed for aspiring drivers by assisting in tracking driving hours. The goal of Milestone is to create a more accessible option for tracking driving hours.

Project Criterea

Functionality

- [x] The ability to add new logs (past log, live log, template log)
- [x] The ability to view past logs
- [x] The ability to edit past logs
- [x] The ability to delete past logs
- [x] The ability to add new templates
- [x] Indicates total amount of time driven
- [x] Indicates total amount of day hours driven
- [x] Indicates total amount of night hours driven
- [x] There are no bugs or issues with Functionality

Presentation

- [x] Professional design (meaning that it look like an app you would find in the app store)
- [x] Ease of use (easy navigation between screens that show you different things. User understands what to do)
- [x] There are no visual defects (it doesn't have inconsistencies in it's appearance such different fonts that don't go together and look bad)

Research

- [x] Research journal that includes a detailed overview of the app and all of its features
- [x] Research journal has dates of when I worked on the app and resources used that day
- [x] Research journal includes a list of resources and explains how they contributed to the development

Terms

Over the course of the journal, I will refer to certain terms through their abbreviations:

- UI: User Interface: the parts of the app that the user interacts with.
- UX: User Experience: how the user experiences the app. This is typically used when designing for ease of use.
- SDK: Software Development Kit: a collection of software that I use to create the application.
- IDE: Integrated Development Environment: software that lets me write code that can be viewed and tested in my app.

Development log

May - June

Preproduction and Planning (Estimated time: 72h)

- Selecting SDK: I plan to use the Flutter SDK because of the programming language: Dart. Dart is very similar to another programming language that I know (C#) so I'm somewhat familiar with the syntax.
- Learning:
  - Dart Programming Language: I learned the dart programming language by creating a small pre-project application. I made a similar version of Milestone with only with basic functions and no UI.
  - Flutter widget types and functions: Resources: https://docs.flutter.dev/
  - Database: For database organization, I'm using the Hive framework. Resources: https://docs.hivedb.dev/ - https://

pub.dev/packages/hive - https://resocoder.com/2019/09/30/hive-flutter-tutorial-lightweight-fast-database/

- Researching UI & UX

June 1, 2023

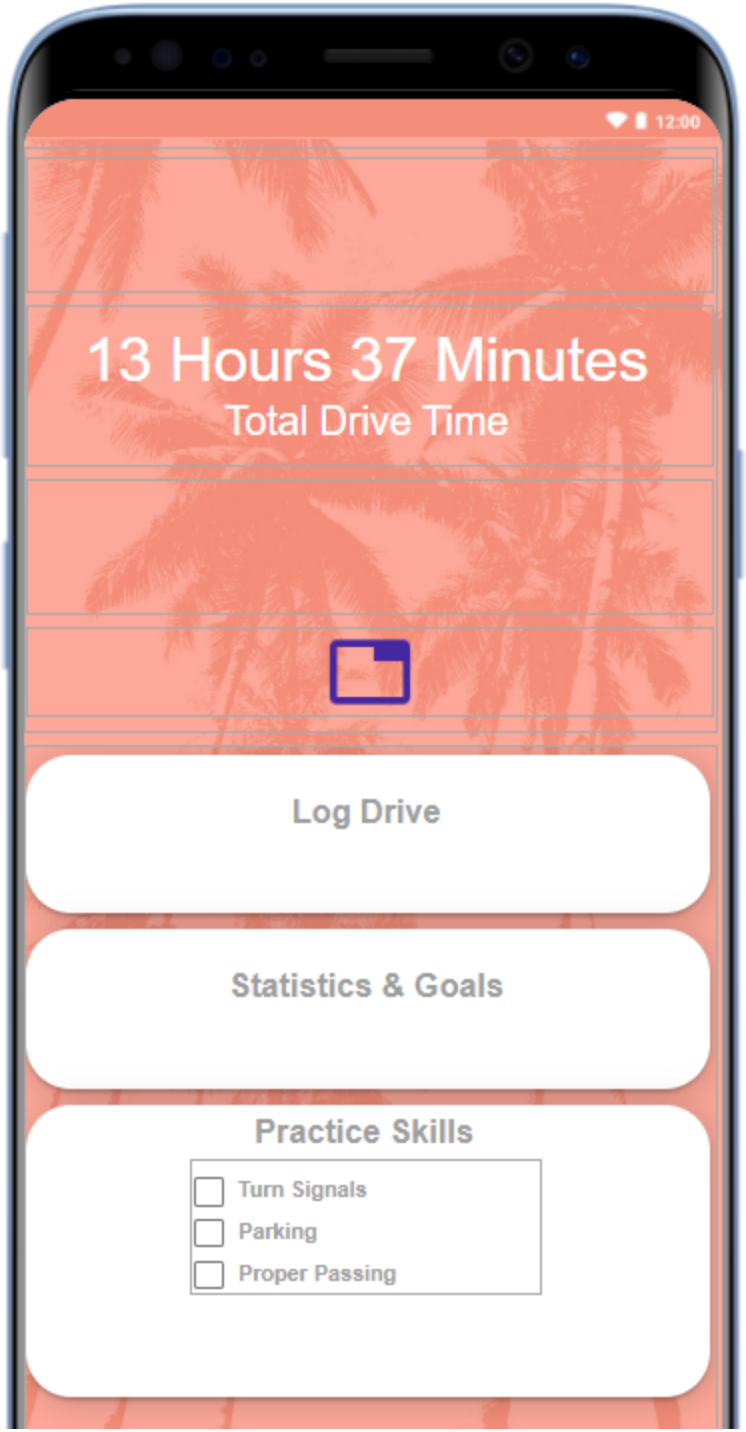
Initial Project Setup (Estimated time: 8h)

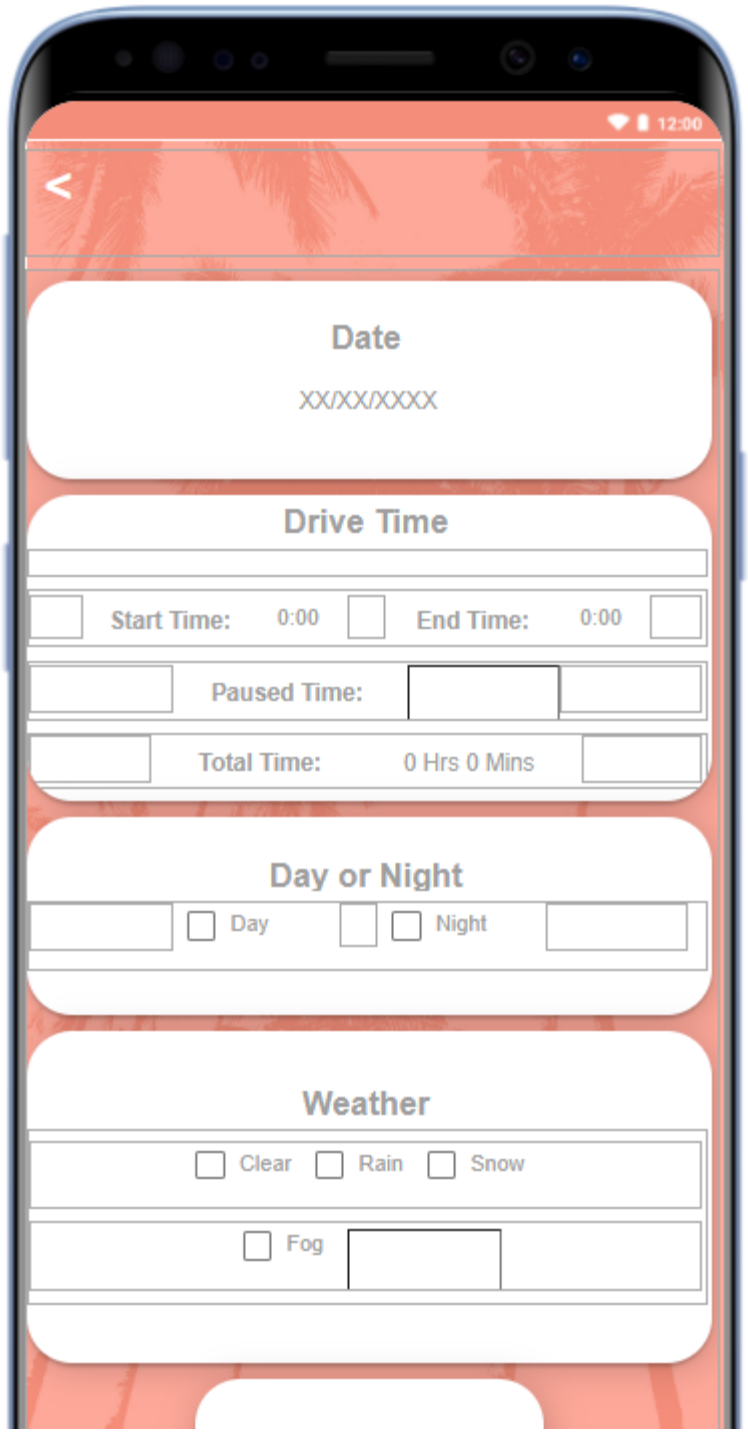
- Created foundational project files, configured the Flutter SDK, initiated a virtual machine for testing, and set up Visual Studio Code IDE. Established the main renderer for UI content display.

June 2 - 4, 2023

Mock UI Creation & UX Planning (Estimated time: 24h)

- Developed a preliminary mock UI system with InVision and engaged in extensive UX planning sessions to map out the user journey.


12:00
13 Hours 37 Minutes
Total Drive Time
Log Drive
Statistics & Goals
Practice Skills
Turn Signals
Parking
Proper Passing


12:00
<
Date
XX/XX/XXXX
Drive Time
Start Time: 0:00
End Time: 0:00
Paused Time:
Total Time: 0 Hrs 0 Mins
Day or Night
Day
Night
Weather
Clear
Rain
Snow
Fog





June 6, 2023

UI Overhaul & UX Research (Estimated time: 10h)

- Conducted a comprehensive UI overhaul and delved into advanced UI and UX research to integrate best practices.

June 7, 2023

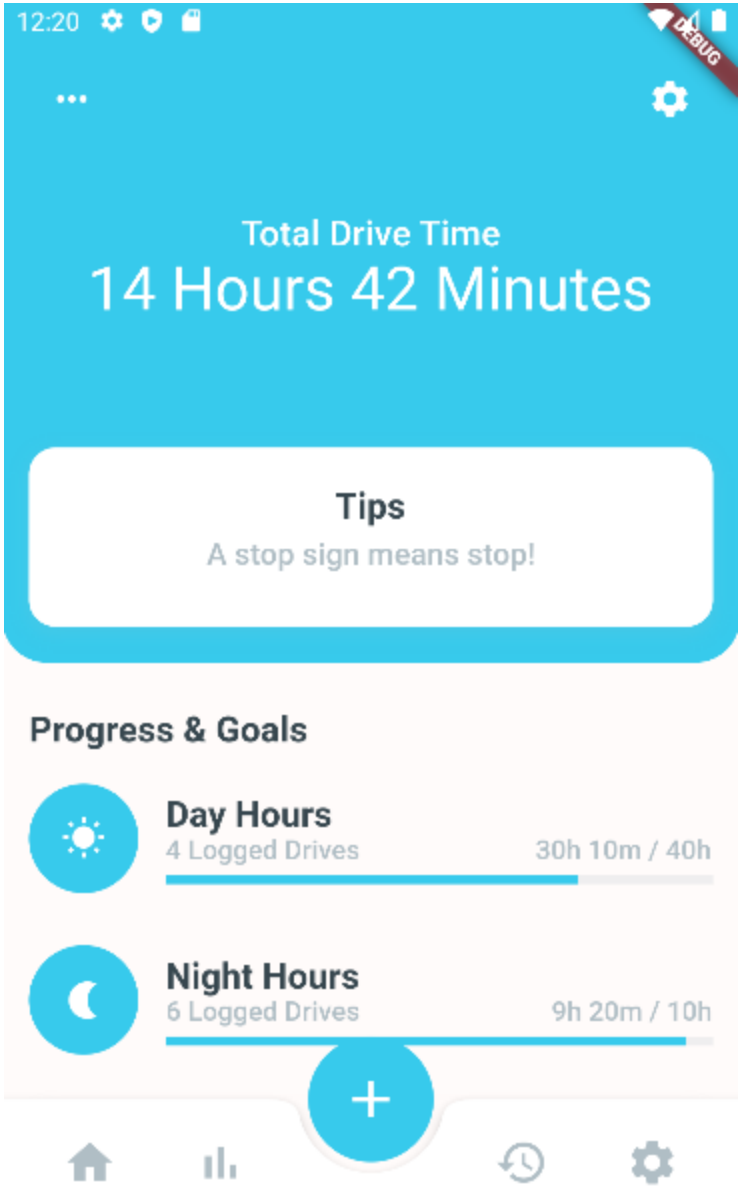
UI Refinement & UX Enhancement (Estimated time: 9h)

- Refined UI components for a polished look and enhanced UX drivers for a responsive user experience.

June 8, 2023

Dashboard Implementation (Estimated time: 8h)

- Implemented the home dashboard, displaying driving statistics and a header for total driving time.



June 9, 2023

Tips Feature & Time Tracking (Estimated time: 7h)

- Introduced a Tips feature and implemented day and night hour tracking for license requirements.

June 10, 2023

Progress Bars & UI Enhancements (Estimated time: 8h)

- Integrated progress bars to visualize achievements and refined the app's color palette.

June 11, 2023

Navigation & Main Button Functionality (Estimated time: 8h)

- Developed a bottom navigation bar and added a main button for new driving session inputs.

June 12, 2023

History Page Development (Estimated time: 9h)

- Created a history page to display past logs with editable and deletable history cards.

June 15, 2023

UI Consistency Check (Estimated time: 6h)

- Reviewed the app's UI for consistency across screens, standardizing fonts, buttons, and colors.

June 20, 2023

Enhanced User Feedback Mechanisms (Estimated time: 7h)

- Improved user action feedback mechanisms with subtle animations and success/error messages.

June 25, 2023

Optimization of Layouts (Estimated time: 8h)

- Refined screen layouts for better content accessibility and readability, focusing on responsive design.

June 30, 2023

Revamped Settings Page (Estimated time: 7h)

- Overhauled the Settings page for more intuitive navigation and grouped settings.

July 5, 2023

Improved Navigation Flow (Estimated time: 7h)

- Refined the app's navigation for a more logical and intuitive user journey.

July 10, 2023

Iconography Update (Estimated time: 6h)

- Updated app icons to align with modern design trends for improved visual appeal.

July 15, 2023

Interactive Elements Enhancement (Estimated time: 6h)

- Enhanced interactive elements like sliders and switches for a more engaging UX.

July 20, 2023

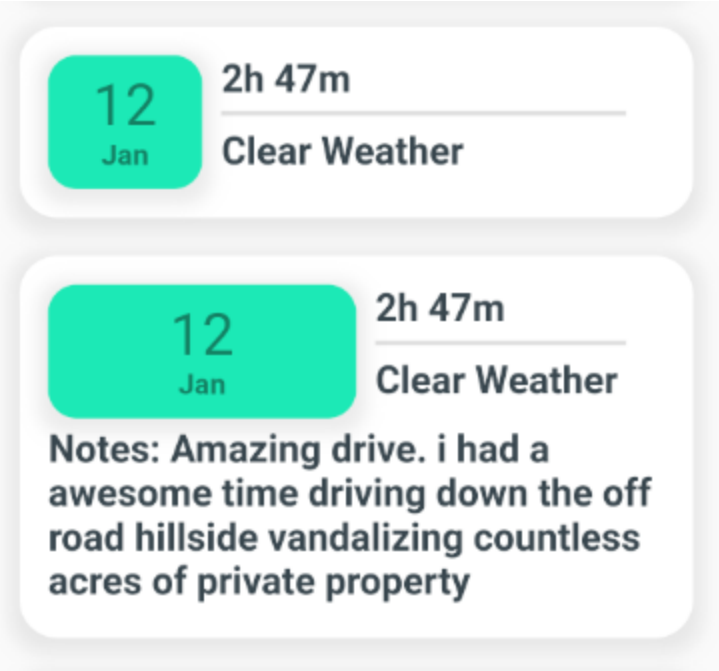
Dashboard Visualization Improvements (Estimated time: 7h)

- Refined dashboard visualizations for clearer statistical information presentation.

July 25, 2023

UI Elements Accessibility Audit (Estimated time: 8h)

- Conducted an accessibility audit of UI elements to ensure broad usability, adjusting as needed.



August 1, 2023

Polishing User Inputs (Estimated time: 8h)

- Refined user input forms and controls for clarity and ease of use, with immediate validation feedback.

August 8, 2023

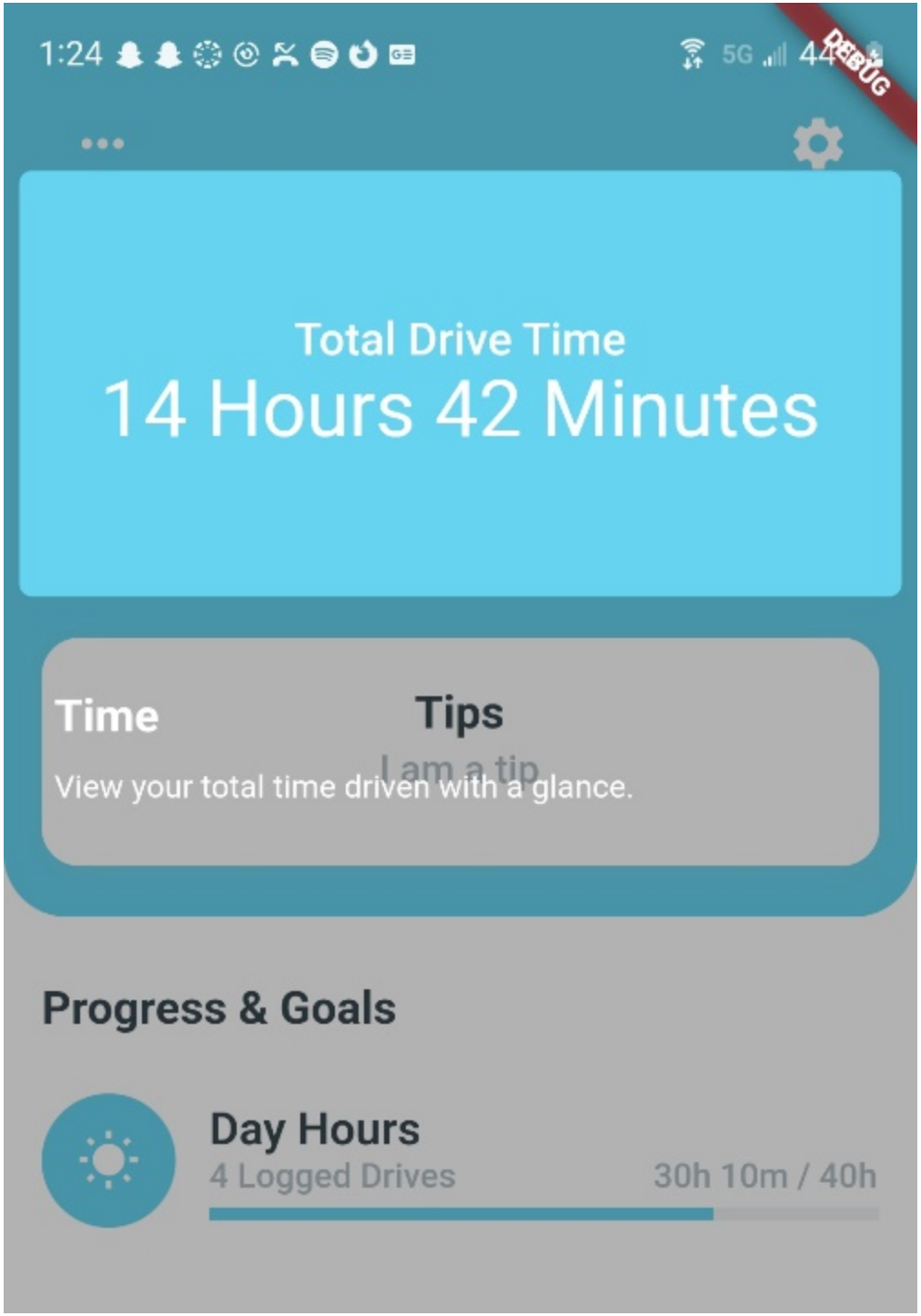
Theme Customization Features (Estimated time: 7h)

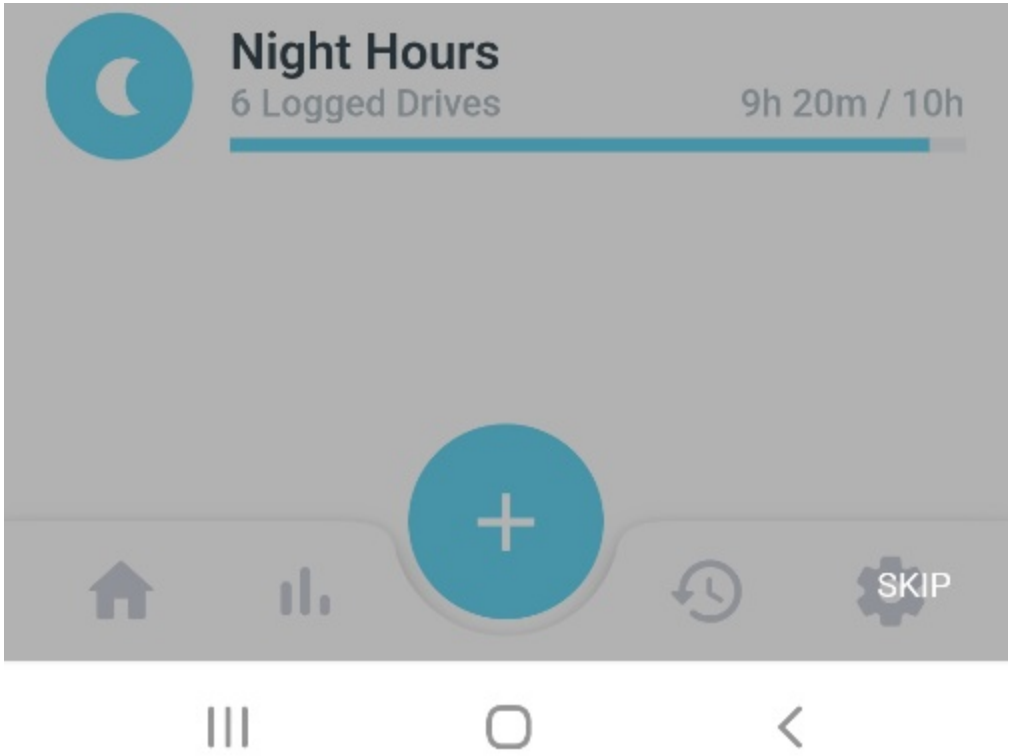
- Introduced theme customization options for user personalization.

September 10, 2023

User Onboarding & Goal Setting (Estimated time: 8h)

- Developed an opening guide for new users and introduced a goal-setting feature.


1:24
5G
DEBUG
Total Drive Time
14 Hours 42 Minutes
Time
Tips
I am a tip
View your total time driven with a glance.
Progress & Goals
Day Hours
4 Logged Drives
30h 10m / 40h



September 11, 2023

Database Recovery & Code Refactoring (Estimated time: 12h)

- Addressed a critical database malfunction with an extensive codebase rewrite for stability.

September 12, 2023

Log Enhancements & Bug Fixes (Estimated time: 8h)

- Enhanced logs with additional details and fixed UI bugs for improved history card rendering.

September 30, 2023

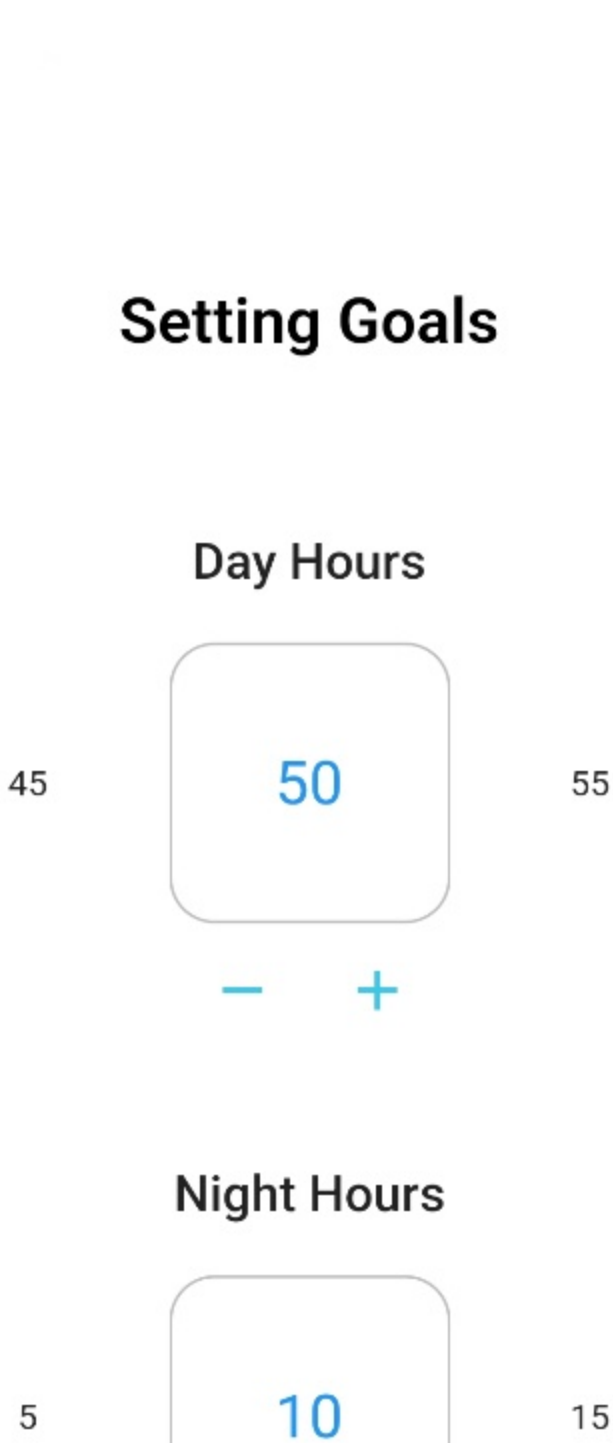
Welcome Screen & User Goals (Estimated time: 9h)

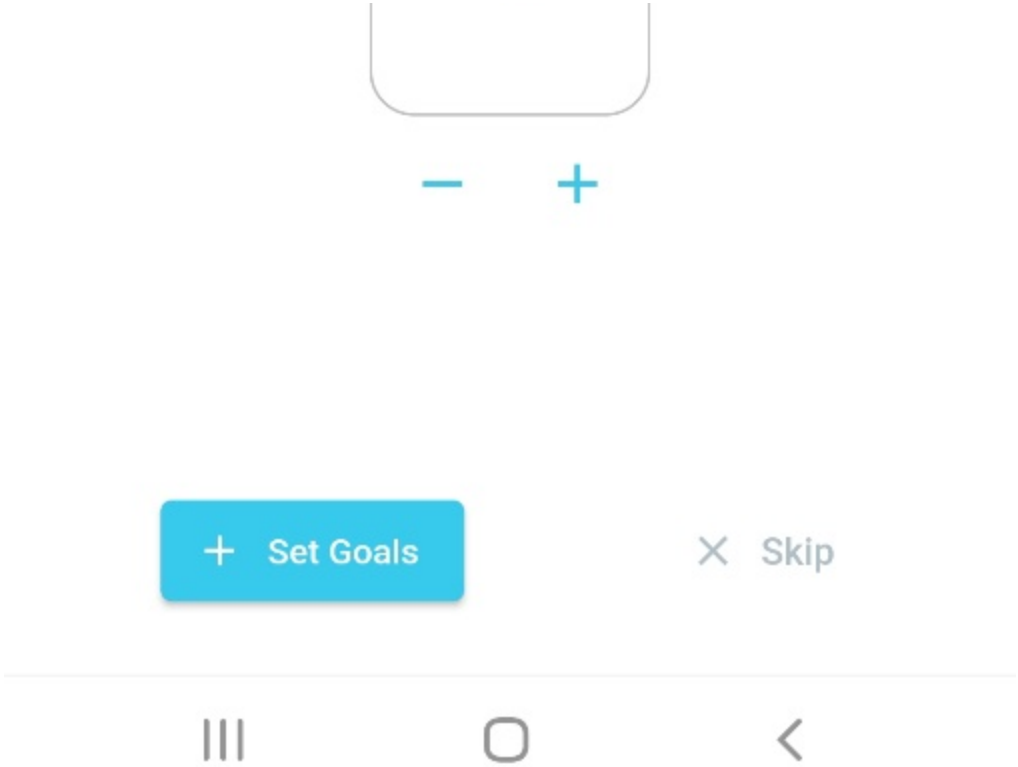
- Enhanced the welcome screen with interactive elements and finalized the goal-setting screen.

September 30, 2023

Welcome Screen & User Goals (Estimated time: 9h)

- Enhanced the welcome screen with interactive elements and finalized the goal-setting screen.


DEBUG
Setting Goals
Day Hours
45
50
55
Night Hours
5
10
15



December 1, 2023

Beta Version Preparation (Estimated time: 8h)

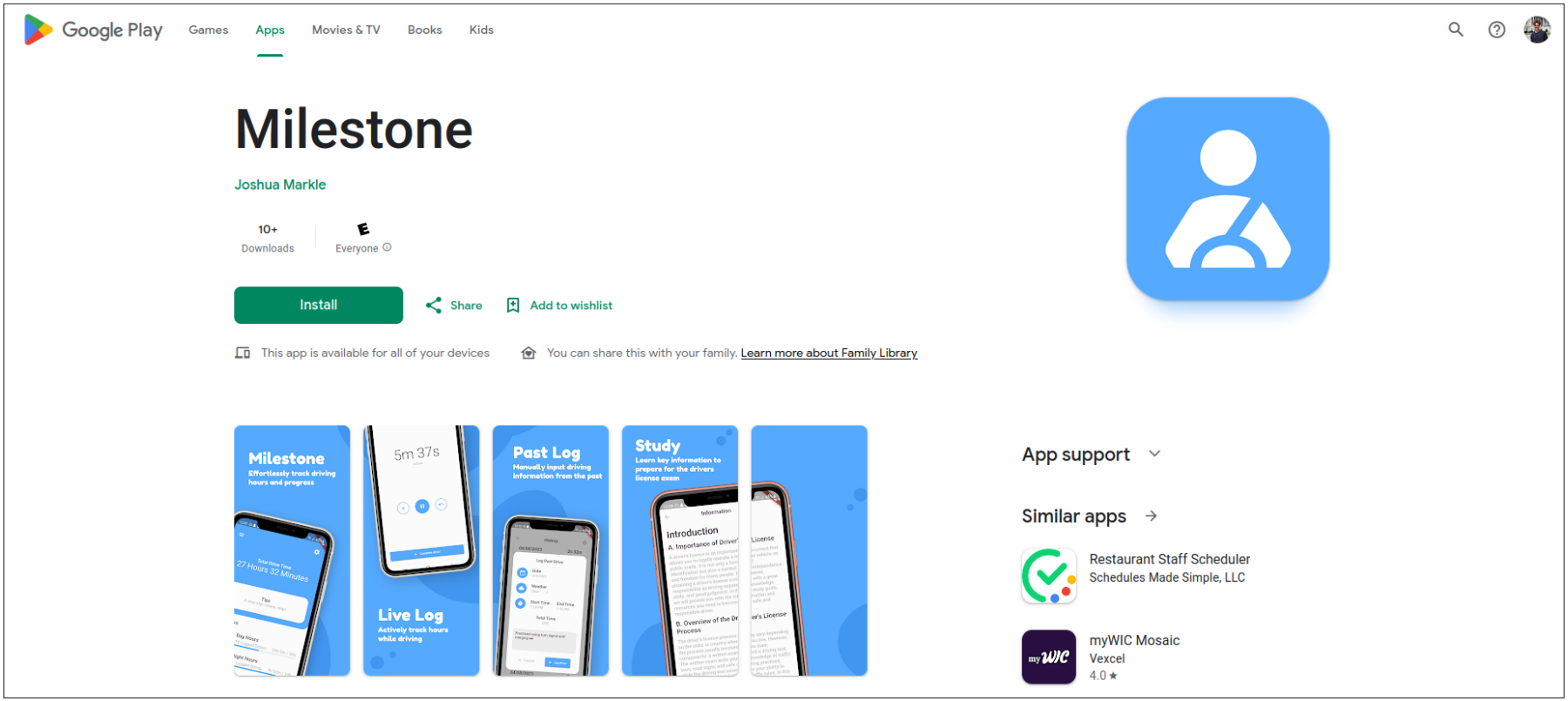
- Initiated preparations for releasing a beta version of the Milestone app.
- Compiled a list of known issues and improvements to address in future updates and set up beta testers.

December 3, 2023

Android Beta Release Setup (Estimated time: 5h)

- Configured the app's settings for the Google Play Store, including setting up a beta testing program.
- Created promotional graphics and materials as required by the Play Store listing.

- Submitted the app for review, and sorting through the Google Play Store guidelines for beta releases.



December 5, 2023

iOS Beta Release Attempt (Estimated time: 2h)

- Attempted to configure the app's settings for the Apple App Store using available online guides and tools.
- Faced significant challenges due to the absence of a Mac, which is essential for using Xcode and completing the iOS app submission process.
- Explored alternative solutions like cloud-based Mac services and borrowing a Mac but encountered issues with app signing and provisioning profiles.

March 1, 2024

Polish The Research Journal (Estimated time: 3h)

- Rewrite the journal using markdown
- Write a small program to compile the markdown into a pdf

March 3, 2024

Complete The Journal (Estimated time: 1h)

- Create a comprehensive list of most of the references used to complete this project

References

Official Documentation & Guides

- **Flutter Official Documentation** (`https://flutter.dev/docs`)
    - The primary resource for all things Flutter, from setup to advanced usage. Used extensively for learning Flutter and troubleshooting during development.
- **Dart Programming Language** (`https://dart.dev/guides`)
    - Official Dart language guides, which were instrumental in understanding Dart syntax and features for Flutter app development.
- **Hive Database Documentation** (`https://docs.hivedb.dev/`)
    - Documentation for Hive, a lightweight and efficient NoSQL database for Flutter. Used for local data storage solutions within the app.

- **Apple Developer Documentation** (`https://developer.apple.com/documentation`)
  - Essential for understanding iOS app development requirements, including guidelines for App Store submissions (attempted for iOS beta release).
- **Google Play Console Help** (`https://support.google.com/googleplay/android-developer`)
  - Guidance on publishing apps on the Google Play Store, managing releases, and setting up beta testing programs.

Tutorials & Articles

- **Flutter & Dart Tutorials by Code With Andrea** (`https://codewithandrea.com/`)
  - Offers practical Flutter and Dart tutorials, which were helpful for specific features like state management and UI design.
- **Flutter State Management Guide** (`https://flutter.dev/docs/development/data-and-backend/state-mgmt/intro`)
  - A comprehensive guide to state management in Flutter, crucial for managing app state and user data effectively.
- **Responsive Design in Flutter** (`https://medium.com/flutter-community/flutter-responsive-design-2e7c9aefb5d1`)
  - An article that provides strategies for creating responsive layouts in Flutter, ensuring the app looks good on various screen sizes.
- **Integrating APIs in Flutter** (`https://flutter.dev/docs/cookbook/networking/fetch-data`)
  - Demonstrates how to fetch data from the internet, useful for any external data or services the app might need to integrate.
- **Testing Flutter Apps** (`https://flutter.dev/docs/testing`)
  - Offers guidelines for writing and running tests in Flutter, ensuring app reliability and helping to catch bugs early.

Community Resources

- **Flutter GitHub Repository** (`https://github.com/flutter/flutter`)
  - The Flutter project on GitHub is a valuable resource for understanding the framework's inner workings and community-contributed issues and solutions.
- **Stack Overflow Flutter Tag** (`https://stackoverflow.com/questions/tagged/flutter`)
  - A community of developers where one can find solutions to specific coding challenges or contribute by answering questions related to Flutter.
- **Flutter Awesome** (`https://flutterawesome.com/`)
  - A curated list of awesome Flutter libraries, tools, tutorials, and articles, useful for discovering packages and resources to enhance the app.
- **Dribbble for UI Inspiration** (`https://dribbble.com/tags/flutter_ui`)
  - A platform for digital design inspiration, where one can find modern UI design ideas and trends applicable to Flutter app development.

Tools & Services

- **InVision App** (`https://www.invisionapp.com/`)
  - A digital product design platform used to create mockups and prototypes of the app's UI, facilitating UX planning and design.
- **GitHub for Version Control** (`https://github.com/`)
  - Utilized for source code management and version control, allowing for efficient collaboration and code history tracking.

- **Pub.dev for Flutter Packages** (`https://pub.dev/`)
    - The official package repository for Flutter, used to find and integrate community-developed packages and plugins into the app.

Note:

The links provided are illustrative and should be replaced with actual URLs when accessing these resources.
